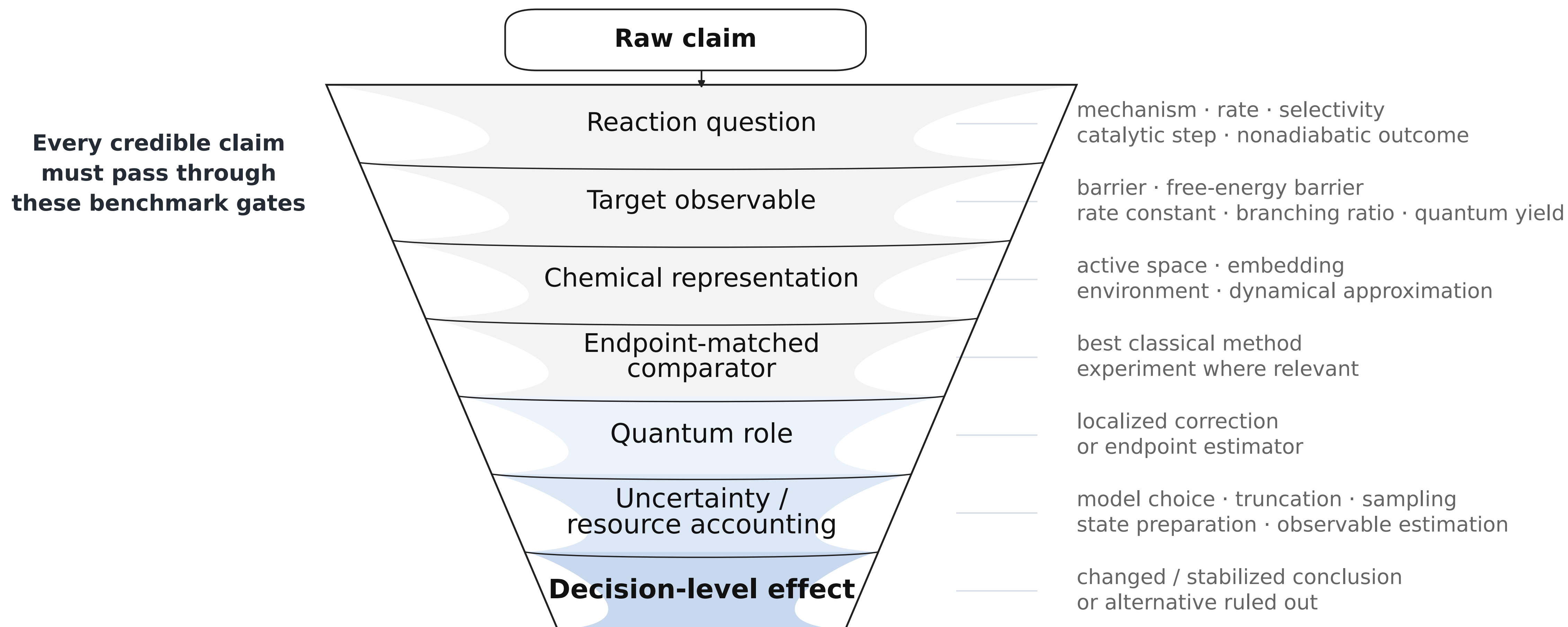


a)

Benchmark funnel



Internal improvements are enabling evidence unless propagated to the reaction observable.

Energy, fidelity, circuit depth, scaling, resource estimates, or an isolated subproblem are not enough.

b)

Localized quantum correction inside a classical workflow

1 classical context

Global reaction workflow remains classical

Reaction hypothesis generation

Structure and ensemble generation

Sampling / environment treatment

Classical screening / free-energy estimation

Kinetic / branching analysis
endpoint propagation → decision

hypotheses, ensembles, environment, screening, free energy, branching, and decision propagation stay classical

2 local bottleneck

Localized quantum correction

Endpoint-controlling bottleneck

Q

where uncertainty controls the endpoint

Electronic structure | Embedding
Dynamics | Estimator / extraction

Endpoint-controlling bottlenecks

embedded active spaces
transition-state regions
surface sites
catalytic active sites
solvated subsystems
multistate / open-system models
spin / correlation ambiguity
estimator-limited observables
nonadiabatic crossings

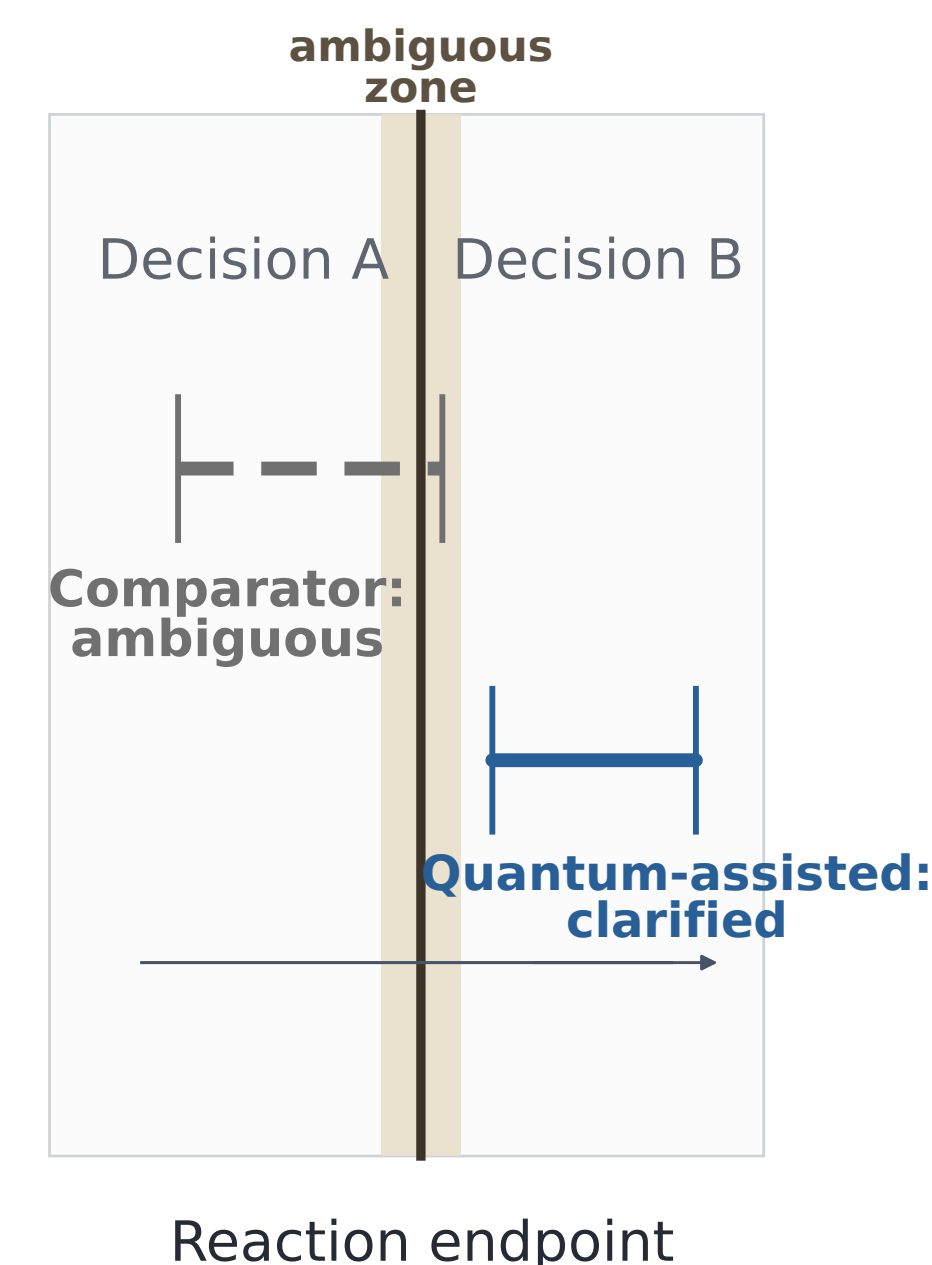
bottleneck selection

endpoint propagation

3 decision test

Reaction decision boundary

schematic decision coordinate



Decision changed, stabilized, or alternative ruled out

Comparator-level ambiguity → localized quantum correction → endpoint propagation → decision-level resolution